

HARDWARE TOOLS SELECTION FOR EMBEDDED SYSTEMS

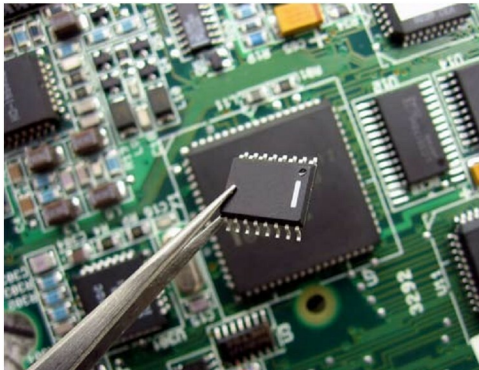
Although most tasks in the design and development of embedded systems have been taken over by software tools, electronic instruments and hardware-based development kits still play an important role in the process—whether as building blocks for the system or as a testing ground for the software and firmware components. What is more, these constantly evolving tools help deliver customer requirements faster and at a lower cost

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Some believe that a computer teamed with the right software can design it all—a car, an airplane, even a spacecraft. However, when designing and developing embedded systems, hardware and software tools always walk in step, helping each other attain perfection. While software helps design the hardware in fast, easy, graphical ways, hardware helps to test the software. Both go hand-in-hand all through the many stages of designing and developing an embedded system, right from hardware platform creation and verification (simulation) to software platform creation, software application creation and software verification (debugging).

Hardware as well as software tools play a critical role in choosing, putting together and testing the various components of an embedded system, viz, microprocessors, microcontrollers or digital signal processors (DSPs), application-specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs), analogue-to-digital and digital-to-analogue converters, communication paths, memory devices and so on.

The key hardware tools used in the development, testing and debugging of embedded systems include



development and evaluation boards, FPGAs, emulators, debuggers, logic analysers and oscilloscopes. These not only help to improve the quality and performance of the targeted embedded system but in many cases also reduce the time-to-market.

As the role of embedded systems in our lives increases constantly—with these little, invisible computers powering devices ranging from microwave ovens to modems—the tools used to make these are also constantly evolving

with the goal of enabling the developers to create more potent embedded systems within a shorter span of time. The hardware tools used for embedded development also fall within this dynamic category.

Today's tools enable faster, cheaper development of even large systems, support new standards and software tools, come pre-packaged for specific purposes, enable easy inclusion of new features such as GPS or high-definition video support in products, and in so